Name of the Student	T.Chaitra
Reg. No	192425217
GitHub Link	https://github.com/chaitramanohar/MLA0402-DEEP-LEARNING

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Mock Interview

COURSE NAME: Deep Learning **COURSE CODE: MLA04**
SLOT :

COURSE OUTCOME COVERED

CO 5: Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

Course Outcome Weightage: 35% **Duration: 90 minutes**
Assessment Tool Weightage: 20% **Mark : 25**

Code of Conduct:
I T.Chaitra(192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.Chaitra **Name of the student:T.Chaitra**

Criteria	Subject Knowledge and Conceptual Understanding (3)	Technical Accuracy and Application of Concepts (2.5)	Problem-Solving and Analytical Thinking **(2)**	Communication Skills and Clarity of Explanation **(1.5)**	Confidence, Presentation and Professional Behaviour ***(1)***	**Total (10)**
Weightage	***30 %***	***25%***	***20%***	***15 %***	***10 %***	**100%**
Q1						
Q2						
Q3						
Q4						
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Q22							
Q23							
Q24							
Q25							

Signature of the Faculty: **Total Marks:**

QUESTIONS: 25 **Total Marks:25**

1.What is an Autoencoder in Deep Learning?
2.What is the main purpose of an Autoencoder?
3.What are the major components of an Autoencoder?
4.What is the role of the Encoder in an Autoencoder?
5.What is the role of the Decoder in an Autoencoder?
6.What is meant by the Bottleneck layer in an Autoencoder?
7.What is meant by Reconstruction in an Autoencoder?
8.How does an Autoencoder learn to reconstruct its input?
9.What is the difference between an Autoencoder and a traditional Neural Network?
10.What type of loss function is commonly used to train an Autoencoder?
11.Explain the basic architecture of an Autoencoder with a suitable example.
12.Why is the dimensionality of the bottleneck layer usually smaller than the input layer?
13.What happens if the bottleneck layer has too many neurons?
14.What happens if the bottleneck layer has very few neurons?
15.What is a Denoising Autoencoder?
16.How can an Autoencoder be used for image denoising?
17.How can Autoencoders be used for dimensionality reduction?
18.How can an Autoencoder be used for anomaly detection?
19.What is the difference between a Denoising Autoencoder and a basic Autoencoder?
20.Give some real-world applications of Autoencoders.
21.What is a Deep Generative Model?
22.What is the difference between a generative model and a discriminative model?
23.What is a Variational Autoencoder (VAE)?
24.What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?
25.What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Subject Knowledge and Conceptual Understanding	30%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Technical Accuracy and Application of Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Problem-Solving and Analytical Thinking	20%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Communication Skills and Clarity of Explanation	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Confidence, Presentation and Professional Behaviour	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

1. What is an Autoencoder in Deep Learning?

An **Autoencoder** is an unsupervised neural network that learns to encode input data into a compact representation and then reconstruct the original input.

2. What is the main purpose of an Autoencoder?

Its main purpose is **feature learning and dimensionality reduction**, and it can also be used for denoising and anomaly detection.

3. What are the major components of an Autoencoder?

The three main components are:
Encoder → Bottleneck/Latent Space → Decoder.

4. What is the role of the Encoder?

The **Encoder** converts the input into a smaller and meaningful latent representation.

5. What is the role of the Decoder?

The **Decoder** takes the latent representation and reconstructs the original input.

6. What is meant by the Bottleneck layer?

The **Bottleneck** is the compressed latent layer between the encoder and decoder. It contains the important features learned from the input.

7. What is Reconstruction?

Reconstruction is the process of generating an output $\hat{x}$that is as close as possible to the original input x.

8. How does an Autoencoder learn to reconstruct its input?

It compares the original input with the reconstructed output using a **reconstruction loss** and updates its weights through **backpropagation**.

9. Difference between Autoencoder and traditional Neural Network?

A traditional neural network usually learns to predict a target label, while an autoencoder generally learns to **reconstruct its own input** without requiring class labels.

10. What loss function is commonly used?

Mean Squared Error (MSE) is commonly used for continuous data, while **Binary Cross-Entropy** can be used for normalized binary-like data.

$$MSE = \frac{1}{n}\sum(x - \hat{x})^2$$

11. Explain the basic architecture with an example.

Example:

Input Image

↓

Encoder

784 → 128 → 32

↓

Bottleneck

32 features

↓

Decoder

32 → 128 → 784

↓

Reconstructed Image

The encoder compresses the image, and the decoder reconstructs it.

12. Why is the bottleneck usually smaller than the input?

A smaller bottleneck forces the network to learn only the **most important features**, enabling dimensionality reduction.

13. What happens if the bottleneck has too many neurons?

The model may simply learn to copy the input, resulting in **poor compression and less meaningful feature learning**.

14. What happens if the bottleneck has very few neurons?

The model may lose important information, resulting in **high reconstruction error**.

15. What is a Denoising Autoencoder?

A **Denoising Autoencoder** is trained using corrupted/noisy input but learns to reconstruct the original clean input.

Noisy Input → Encoder → Decoder → Clean Output

16. How can an Autoencoder be used for image denoising?

Add noise to clean images, train the autoencoder to reconstruct the clean images, and then use the trained model to remove noise from new images.

17. How can Autoencoders be used for dimensionality reduction?

The encoder maps high-dimensional input into a smaller **latent representation**, which can then be used as reduced-dimensional features.

18. How can an Autoencoder be used for anomaly detection?

Train the autoencoder mainly on normal data. Anomalies usually produce **high reconstruction error**, so a threshold can be used to identify them.

Low Error → Normal

High Error → Anomaly

19. Difference between Denoising Autoencoder and basic Autoencoder?

Basic AE	Denoising AE
Clean input → reconstruction	Noisy input → clean reconstruction
Learns reconstruction	Learns robust reconstruction
Less focused on noise	Designed to handle noise

20. Give real-world applications.

Common applications include:

- Image denoising
- Dimensionality reduction
- Anomaly detection
- Feature extraction
- Data compression
- Fraud detection
- Medical-image analysis

21. What is a Deep Generative Model?

A **Deep Generative Model** is a deep learning model that learns the underlying data distribution and can generate new samples similar to the training data.

Examples: **VAE, GAN, DBN and DBM.**

22. Difference between Generative and Discriminative models?

Generative	Discriminative
Learns data distribution	Learns decision boundary
Can generate new data	Mainly predicts labels
Example: GAN, VAE	Example: Logistic Regression, CNN classifier

Easy memory:
Generative = Generate
Discriminative = Decide

23. What is a Variational Autoencoder (VAE)?

A **VAE** is a generative autoencoder that learns a **probability distribution** in the latent space and samples from it to generate new data.

Input

↓

Encoder

↓

μ, σ

↓

Sampling

↓

Latent z

↓

Decoder

↓

Generated/Reconstructed Output

24. Difference between traditional Autoencoder and VAE?

Traditional AE	**VAE**
Deterministic latent representation	Probabilistic latent distribution
Mainly reconstruction	Reconstruction + generation
No explicit distribution constraint	Uses a structured latent distribution
Less suitable for controlled generation	Suitable for generating new samples

VAE loss:

$$L = L_{reconstruction} + L_{KL}$$

25. What are GANs? Roles of Generator and Discriminator?

A **Generative Adversarial Network (GAN)** consists of two neural networks that compete with each other.

Generator: Creates fake/synthetic data.

Discriminator: Determines whether the data is **real or fake**.